Master's Thesis Exposé

Sim-to-Real Transfer in Mobile Manipulation: Drift Accumulation Analysis for Autonomous Box Stacking on the LIMO Cobot

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Title

Primary Title:
Sim-to-Real Transfer in Mobile Manipulation: Drift Accumulation Analysis for Autonomous Box Stacking on the LIMO Cobot

Alternative titles considered:

- Autonomous Mobile Manipulation for On-Demand Manufacturing: A Sim-to-Real Study on the LIMO Cobot
- Iterative Pick-and-Stack with a Compact Mobile Manipulator: Quantifying Pose Drift and Sim-to-Real Gap
- Coordinated Navigation and Manipulation on the Agilex LIMO Cobot: From Gazebo Simulation to Physical Deployment

Research Questions

Primary Research Question:

Can a compact **differential-drive** mobile manipulator (Agilex LIMO Cobot) autonomously perform a multi-cycle vertical box-stacking task, and how does accumulated navigation and manipulation error constrain the number of achievable stacking cycles before grasp failure?

Sub-Questions

1. **Pipeline integration.** How should the ROS 2 navigation stack (Nav2) and manipulation framework (MoveIt 2) be configured and coordinated for a *multi-step* pick-transport-place-return cycle on a single resource-constrained platform?
2. **Drift characterisation.** How does position error accumulate over repeated navigation cycles, and at what drift magnitude does the arm's reachable envelope fail to encompass the target object?
3. **Simulation-to-reality transfer.** What is the quantitative gap between the simulation baseline (perfect odometry, zero drift) and real-hardware performance — which subsystems transfer directly and where does the gap emerge?

Objectives

1. **Pipeline Integration** — Design and implement a complete multi-cycle mobile manipulation pipeline integrating Nav2 (differential-drive navigation), MoveIt 2 (collision-aware arm planning), and HSV+depth-camera perception, coordinated by a mission state machine that executes *pick* → *transport* → *stack* → *return* cycles without human intervention.
2. **Stacking Experiment** — Implement a vertical box-stacking experiment in Gazebo: the robot picks boxes from a source table one by one, navigates to a target table, places each box on top of the previous one, and returns for the next box. Run $N = 20$ iterations.
3. **Drift Quantification** — Measure per-iteration navigation position error and grasp success rate. Identify the iteration N^* at which drift causes the first grasp failure. Characterise the relationship between accumulated drift and task failure mode.
4. **Sim-to-Real Evaluation** — Deploy the validated pipeline on the physical LIMO Cobot. Compare simulation drift profiles (upper bound: perfect odom) against hardware drift profiles (real encoder odometry + AMCL) to quantify the sim-to-real gap and identify which assumptions the simulation makes that break on hardware.

Scope and Limitations

In Scope

- ✓ Autonomous navigation in a controlled, static indoor environment (flat ground, known layout, static obstacles).
- ✓ **Differential-drive** kinematic model — permits in-place rotation, enabling the robot to align precisely with either table before each pick/place.
- ✓ Multi-cycle vertical box stacking: pick from Table 1, place on Table 2, repeat for up to $N = 20$ cycles.
- ✓ 6-DoF arm manipulation of small standardised boxes (known colour, known size).
- ✓ Depth-camera object detection and 3-D pose estimation (HSV + pin-hole de-projection).
- ✓ Quantitative drift analysis: per-iteration position error and grasp success.
- ✓ Sim-to-real comparison: Gazebo (perfect odom) vs. physical hardware (AMCL + real encoders).

Out of Scope

This thesis does NOT include:

- Dynamic or outdoor environments.
- Dynamic obstacle avoidance during manipulation.
- Multi-robot coordination or fleet management.
- Advanced or learned grasping strategies (no deep learning).
- Perception of unknown or arbitrary objects (predefined coloured boxes only).
- Ackermann steering — this kinematic mode was prototyped and abandoned in favour of differential drive (which permits in-place rotation).

Limitations (Honest Assessment)

- **Simulation odometry is perfect.** The Gazebo baseline uses `odometry_source: WORLD`, which produces drift-free pose estimates. This is intentional — it establishes the upper performance bound. Hardware trials will introduce real drift.
- **Depth camera blind spot.** The Orbbec DaBai has a physical minimum sensing distance of 0.30 m, which is larger than the grasp distance (≈0.21 m). This is a known constraint treated as a sim-to-real finding rather than a blocker.
- **Arm reach limits stacking height.** The myCobot 280 has a 280 mm nominal reach; for top-down grasps the effective reach is shorter (approx. 180 mm forward). A growing stack eventually leaves the arm's reachable envelope. This physical limit defines the maximum achievable stack height independently of drift.
- **Weld model.** Grasping in simulation uses a rigid body weld. This tests the navigation, perception, and planning layers but does not validate frictional grasp closure on the physical gripper.

Overview and Context

Autonomous mobile manipulation — the integration of autonomous navigation with robotic arm manipulation on a single platform — is a core capability for on-demand manufacturing. In smart manufacturing scenarios such as the RoboCup Smart Manufacturing League's Embodied Autonomous Intelligence Workshop (EAI-WS) [**?**], robots must navigate shared spaces, locate and grasp target objects, transport them, and place them precisely at target locations — a complete *pick-transport-place* cycle that must execute reliably over repeated iterations.

This thesis investigates this task on the **Agilex LIMO Cobot**: a compact mobile manipulator combining a differential-drive mobile base (LIMO PRO, 4 kg payload, 1 m/s maximum speed) with a 6-DoF robotic arm (Elephant Robotics myCobot 280, 250 g payload, 280 mm nominal reach), an Orbbec DaBai depth camera, an EAI T-mini Pro LiDAR, and an NVIDIA Jetson Orin Nano compute unit. The system is built within the **ROS 2 Humble** ecosystem using **Nav2** for autonomous navigation (differential-drive, AMCL localisation, SLAM Toolbox mapping), **MoveIt 2** for collision-aware arm motion planning, and **Gazebo Classic** for simulation.

The experimental scenario is a **multi-cycle vertical box-stacking task**: the robot picks boxes one at a time from a source table (Table 1), navigates to a target table (Table 2), places each box on top of the previous one (building a vertical stack), returns to Table 1, and repeats — up to 20 cycles. This scenario is chosen because it *exercises all subsystems under compounding error*: each cycle's starting pose depends on the navigation accuracy of the return leg, so any odometry drift or localisation error accumulates and eventually exceeds the arm's grasp tolerance. The iteration at which the first grasp failure occurs is a concrete, platform-specific metric that characterises the system's operational limits.

A central contribution is the **sim-to-real analysis**: the full 20-cycle experiment is run in simulation (establishing an upper bound with perfect odometry), and then on the physical hardware (exposing real drift), allowing a direct quantitative comparison of how well simulation predicts real performance.

Completed Work (As of June 2026)

The simulation baseline is **fully operational**. Every subsystem listed below has been implemented, debugged, and verified in Gazebo. 29 engineering faults were identified and resolved.

- ✓ **Verified platform model.** Combined Xacro model with audited physical parameters: wheel track 0.172 m (confirmed from datasheet), DaBai H-FOV 67.9°, LiDAR ±120°/0.10–12 m (forward arc; self-hit filtering), gripper jaw range 20–45 mm.
- ✓ **Navigation pipeline.** Nav2 + SLAM Toolbox + AMCL on a differential- drive base. Clean occupancy map built (238×358 cells, 0 noise pixels, 97.2% free space). Costmap tuned for arm self-masking (obstacle min range raised to 0.35 m). Robot navigates autonomously to any goal in the workspace.
- ✓ **Perception pipeline.** HSV centroid detection + depth de-projection via pin-hole model. Publishes 3-D `PoseStamped` on `/box_pose`. Verified: correct pose estimated within grasp range.
- ✓ **Arm manipulation.** MoveIt 2 with 22-seed KDL IK and RRTConnect. Reachability mapped: 25.9% overall, top-down reach contracts sharply at $x > 0.18$ m. Full grasp sequence (ready → pre-grasp → descent → close → lift) executing correctly. Calibrated: lateral offset, orientation tolerance 0.1 rad, closing clearance.
- ✓ **Integrated nav→pick→backup pipeline.** Single orchestrator node sequences travel posture → navigate → visual dock → grasp → backup. Travel posture implemented to suppress joint-2 pendulum oscillation during navigation. Live TF weld service (stale-pose bug fixed). Demonstrated in simulation.
- ✓ **Simulation environment hardened.** GPU PRIME offload configured for NVIDIA RTX 3060 (eliminates depth-camera segfault on AMD iGPU). Gazebo physics stabilised (removed `contact_max_correcting_vel=0.1` instability).
- ✓ **Thesis chapters drafted.** Chapters 1, 3, 4, 5, 6 written with real results and detailed engineering fault catalogue (29 entries). Chapter 2 (literature review) is the remaining writing gap.

Remaining Engineering Work

1. **Place stage** (currently a stub) — implement arm motion to lower box onto the stack and release.
2. **Two-table navigation** — add Table 2 to world, add return waypoint.
3. **Stacking height tracking** — increment Z-target by one box height per cycle.
4. **Iteration loop + data logger** — run 20 cycles, log per-iteration pick/place success, nav position error, detected box pose.
5. **Hardware deployment** — swap Gazebo plugins for real hardware drivers, run 20 cycles on physical LIMO Cobot.

Gantt Chart: Remaining Schedule

Reference date: 2026-06-24. Target submission: 2026-09-15.

All simulation subsystems are complete. Remaining work: place stage, stacking experiment, hardware trials, analysis, and writing.

Wk	Focus	Calendar (2026)											
		Jun 24	Jul 1	Jul 8	Jul 15	Jul 22	Aug 29	Aug 5	Aug 12	Aug 19	Sep 26	Sep 2	Sep 9
1	Place stage + world	•											
2	Stacking loop + logger		•										
3	Sim experiment (20 runs)			•									
4	Data analysis + plots				•								
5	Literature review (Ch. 2)					•							
6	Hardware bringup (nav)						•						
7	Hardware stacking trials							•					
8	Sim-to-real analysis								•				
9	Ch. 3–4 finalise									•			
10	Ch. 5–6 results/conc.										•		
11	Full draft + supervisor											•	
12	Corrections + submit												•

Table 1: *

• Done • In progress (current week) • Planned

Milestones

Milestone	Target Date	Deliverable
`m1-sim-complete`	*Done (Jun 2026)*	Full pick→nav pipeline working in sim.
`m2-place-stage`	Jul 7	Place stage + two-table nav working in sim.
`m3-stacking-sim`	Jul 21	20-iteration stacking experiment in sim; CSV data.
`m4-literature`	Aug 4	Chapter 2 (literature review) complete draft.
`m5-hardware`	Aug 18	20-iteration stacking on real LIMO Cobot; CSV data.
`m6-analysis`	Sep 1	Sim-to-real drift comparison plots; Ch. 5 results.
`thesis-draft-1`	Sep 8	Complete first full thesis draft for supervisor review.
`thesis-submit`	Sep 15	Final corrected thesis submitted.

Risks and Mitigation

Risk	Mitigation
Stack height exceeds arm reach before 20 cycles.	The arm reach limit is a physical finding, not a failure. Report it as the geometric upper bound on stack height, separate from drift-induced failure.
Real hardware odometry drift is too large to stack even 2–3 boxes.	This is itself a result: document the minimum drift condition, propose AMCL re-localisation triggers between cycles as future work.
Depth camera blind spot (<0.30 m) prevents box detection at grasp range on hardware.	Treat as sim-to-real finding. Mitigate by approaching to a slightly larger standoff, using remembered pose from farther detection.
Hardware time is limited (lab access).	Stage deployment: navigation first, then arm. Partial hardware results are still valid sim-to-real data.
Literature review delayed (currently a gap).	Allocate Week 5 exclusively to Ch. 2. Use the existing annotated reading list in `docs/references/reading_list.md` as starting material.

References

- Macenski, S. and Jambrecic, I., "SLAM Toolbox: SLAM for the Dynamic World," *JOSS*, 2021.
- Macenski, S. et al., "The Marathon 2: A Navigation System," *IROS*, 2020.
- Masannek, M. et al., "Embodied Autonomous Intelligence for On-Demand Manufacturing," EAI-WS Proposal, 2026.
- Dissanayaka, S. et al., "From Production Logistics to Smart Manufacturing: The Vision for a New RoboCup Industrial League," 2025.

Full bibliography maintained in `docs/thesis/bibliography.bib`.